# Architecture-Aware Diagnosis of Parameter-Efficient Adaptation in Prithvi-100M: Fused-QKV Failure, Capacity Boundaries, and Cross-Domain Remote-Sensing Validation

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**Abstract**

Parameter-efficient fine-tuning (PEFT) is widely used to adapt large pre-trained models, but its behavior on geospatial foundation models is easily confounded by sensor mismatch, spectral-channel bridging, domain shift, and architecture-specific implementation details. This paper studies these confounders for Prithvi-100M. We evaluate seven adaptation strategies across five heterogeneous input configurations on EuroSAT, validate the main ranking on BigEarthNet-S2, extend the protocol to LandCover.ai semantic segmentation, and test whether the ranking survives a production-style Linhe County land-use/land-cover workflow and LoveDA urban/rural cross-domain transfer. The study covers 127 public-benchmark runs, a 35,920-patch Linhe corpus supervised by Esri-derived land-cover labels, a 30-run LoveDA cross-domain sweep, LoveDA full fine-tuning baselines, a completed EuroSAT channel-bridge ablation comparing deterministic zero padding with a learned 10-to-6 channel projection, and an 18-run SatMAE-compatible EuroSAT second-backbone audit.

The results support a diagnostic rather than universal conclusion. First, LoRA underperformance on Prithvi-100M has two separable causes: standard module-wise insertion misses the packed query, key, and value projections in fused-QKV attention, and corrected split-QKV LoRA still saturates near 0.707 OA even when its trainable parameter budget is increased to 4.73M. Second, Houlsby adapters provide the most reliable adaptation among the evaluated PEFT methods, reaching 94.5% of the full fine-tuning score on EuroSAT s2_full while training 1.4% of the parameters in the original benchmark and rising to 0.901 OA in the completed capacity sweep. Third, input modality and domain shift strongly condition PEFT behavior: learned channel bridging improves EuroSAT accuracy while preserving the Houlsby-over-linear ordering, the SatMAE-compatible second-backbone audit preserves the Houlsby-over-LoRA ordering on EuroSAT, and LoveDA cross-domain diagnostics suggest that sub-million-parameter adapters can recover weak binary structure without recovering full multi-class semantic transfer.

These findings do not establish a universal PEFT ranking for all geospatial foundation models. They show that architecture-aware inspection, channel-bridge controls, second-backbone checks, and parameter-capacity audits are necessary before transferring PEFT intuitions from mainstream vision to remote-sensing backbones. The resulting benchmark is therefore both a practical guide for Prithvi-100M deployment and a reproducible stress-test

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protocol for future GeoFM adaptation studies.

*Keywords:* geospatial foundation models, parameter-efficient fine-tuning, Prithvi-100M, remote sensing benchmark, semantic segmentation, domain shift, land-use/land-cover mapping

## 1. Introduction

Geospatial foundation models (GeoFMs) have recently emerged as a promising paradigm for Earth observation analysis. Models such as Prithvi-100M Jakubik et al. (2023), SatMAE Cong et al. (2022), Scale-MAE Reed et al. (2023), and related self-supervised transformers Hong et al. (2024) suggest that large-scale pre-training on satellite imagery can yield transferable representations for downstream remote sensing tasks. At the same time, practitioners increasingly face adaptation problems that differ substantially from the homogeneous settings used during pre-training. Real-world inputs may contain different band subsets, missing channels, alternative sensor combinations, or strongly out-of-distribution modality configurations.

In mainstream natural language processing and computer vision, parameter-efficient fine-tuning (PEFT) methods are widely used to adapt large pre-trained models at low cost. LoRA Hu et al. (2022), adapters Houlsby et al. (2019), BitFit Ben Zaken et al. (2022), and prompt-style tuning methods Lester et al. (2021); Jia et al. (2022) have become standard choices because they often approach full fine-tuning performance with far fewer trainable parameters. This success naturally raises the question of whether the same PEFT rankings hold for geospatial foundation models.

We argue that this assumption is non-trivial for at least three reasons. First, geospatial foundation models often operate on heterogeneous multi-band inputs rather than fixed RGB inputs. Second, remote sensing pre-training objectives, especially masked autoencoding on spectral-spatial data, differ from the language modeling and image classification setups where PEFT methods are most established. Third, implementation details that are harmless in conventional vision backbones can become critical in GeoFMs, especially when architectural components such as fused attention projections interact with adaptation layers.

Despite the practical importance of this problem, current literature lacks a systematic benchmark of PEFT methods under heterogeneous geospatial input settings. Existing studies often evaluate only one or two adaptation strategies, focus on a single downstream task, or treat PEFT as an implementation detail rather than the object of study. As a result, practitioners still lack clear guidance on which PEFT method to choose when adapting a model such as Prithvi-100M to non-standard remote sensing inputs.

A second, related gap is that PEFT comparisons are almost always reported on curated public datasets. It is rarely demonstrated that the same ranking reproduces on operational GIS pipelines, where imagery is heterogeneous in sensor and resolution, labels are derived from third-party products of varying quality, and validation splits must respect spatial structure to prevent leakage. The absence of such a demonstration is a real obstacle for practitioners deciding whether to ship a PEFT-adapted GeoFM into a production workflow. A third gap, complementary to the first two, is that within-domain PEFT comparisons cannot tell a practitioner what to expect when the target domain drifts substantially from pre-training. Cross-domain protocols on remote-sensing benchmarks — where train and test are drawn from explicitly disjoint imaging conditions, geographies, or sensor families — are routine in the dataset literature but largely unstudied for PEFT method ranking on geospatial foundation models.

This paper addresses all three gaps through a controlled empirical study on Prithvi-100M, paired with a production-style validation and a public cross-domain replication. We benchmark seven adaptation strategies across five modality configurations on EuroSAT, validate the main findings on BigEarthNet-S2, extend the protocol to semantic segmentation on LandCover.ai, re-instantiate the resulting five-method comparison on a 35,920-patch Linhe County corpus collected from 203 Chinese-domestic high-resolution scenes (0.42–4.29 m GSD, 22 satellite families) with Esri 2022 global LULC used as supervisory labels, and finally replicate the same five-method protocol on the LoveDA urban/rural cross-domain benchmark. Beyond reporting final scores, we include a full fine-tuning ceiling, a diagnostic split-QKV LoRA ablation, a split-QKV rank-sensitivity audit, a completed EuroSAT channel-bridge ablation, an 18-run SatMAE-compatible EuroSAT second-backbone audit, a synthetic weak-label control on the Linhe corpus, and a per-class diagnostic on the LoveDA cross-domain split. The Linhe control is not an independent manual validation set; it tests whether the measured adapter delta changes when the same imagery is paired with a deliberately simple weak label. Across these experiments, we show that PEFT rankings commonly assumed in mainstream NLP and vision should not be transferred to Prithvi-100M without architecture-aware validation. We also show that benchmark rankings can be stress-tested under a geographic-split production-style workflow, while cross-domain segmentation exposes capacity and label-source limits that are not visible in within-domain classification alone.

Our main contributions are as follows:

1. We report a systematic benchmark of PEFT methods on Prithvi-100M under heterogeneous input settings and add a SatMAE-compatible EuroSAT second-backbone audit, covering 127 public-benchmark experiments across EuroSAT, BigEarthNet-S2, LandCover.ai, and the second-backbone validation.

2. We separate implementation failure from post-fix capacity limits for LoRA on a fused-QKV Prithvi-100M backbone: standard module-wise insertion can miss packed attention projections, while corrected split-QKV low-rank adaptation still saturates below bottleneck adapters in the tested settings.

3. We introduce an architecture-aware PEFT audit protocol that records trainable tensors, tests fused-attention insertion, checks channel-bridge sensitivity, and compares low-rank capacity against bottleneck-adapter capacity before interpreting negative PEFT results.

4. We show that Houlsby adapters are the strongest method in our Prithvi-100M benchmark, achieving 94.5% of full fine-tuning performance with only 1.4% of the trainable parameters on EuroSAT and a +155% relative mIoU gain on dense prediction.

5. We confirm across datasets that modality selection has a larger practical impact than PEFT selection: richer spectral inputs consistently outperform reduced-band alternatives.

6. We provide a bounded result for standalone input-stage adaptation (GeoAdapter): it is unstable on public classification benchmarks near Prithvi's pretraining modality, but becomes useful on RGB-only production-style inputs where the modality gap is larger.

7. We test the benchmark ranking on a Linhe County LULC workflow under a geographic scene-level split, recovering Houlsby +0.116 mIoU over linear probing with Esri-derived supervisory labels; we also provide a synthetic weak-label control showing that the measured PEFT delta is sensitive to label source, while noting that this is not an independent manual validation set.

8. We replicate the five-method ranking on LoveDA urban/rural cross-domain segmentation across 30 runs and report evidence consistent with an adapter-capacity threshold under domain shift. We treat this threshold as a hypothesis supported by the present diagnostics, not as a universal rule for all GeoFMs.

9. We complete a SatMAE-compatible second-backbone EuroSAT audit showing that Houlsby remains top-ranked over linear probing and split-QKV LoRA on both s2_full and rgb, converting the former single-backbone limitation into bounded two-backbone evidence.

10. We release a reproducible Geo-MLOps benchmarking framework together with the AlphaEarth-System platform that orchestrates the Linhe deployment, including its data pipeline, training engine, and web dashboard.

## 2. Related Work

### *2.1. Geospatial Foundation Models*

Recent years have seen rapid growth in foundation models for Earth observation. Prithvi-100M Jakubik et al. (2023), SatMAE Cong et al. (2022), Scale-MAE Reed et al. (2023), and SpectralGPT Hong et al. (2024) all extend the masked autoencoder paradigm He et al. (2022b) from natural images to multi-spectral and multi-temporal satellite imagery. These models share the Vision Transformer backbone Dosovitskiy et al. (2021) but differ from conventional vision encoders in three important ways: they ingest multi-spectral inputs rather than RGB, they exploit large unlabeled satellite corpora through self-supervised pre-training Mañas et al. (2021); Wang et al. (2023), and they must support broad variation in spatial resolution and sensor configuration. While downstream transfer results are encouraging, the question of *how* to adapt such models efficiently to new sensor combinations has received much less attention than the pre-training itself.

### *2.2. Parameter-Efficient Fine-Tuning*

Parameter-efficient fine-tuning (PEFT) has become the standard recipe for adapting large pre-trained models without updating all parameters. BitFit Ben Zaken et al. (2022) restricts learning to bias terms; LoRA Hu et al. (2022) injects low-rank updates into linear projections; Houlsby adapters Houlsby et al. (2019) insert bottleneck modules between transformer blocks; and prompt-based methods Lester et al. (2021); Jia et al. (2022) prepend learnable tokens to the input. Subsequent work has unified these strategies under a common framework He et al. (2022a) and explored composition across tasks Pfeiffer et al. (2020). In language models and conventional vision transformers, these methods often approach full fine-tuning while reducing memory and storage costs by one to three orders of magnitude. However, their relative ranking depends strongly on architecture, task, and pre-training objective, which raises the possibility that PEFT rankings established in mainstream domains may not transfer to geospatial models.

### *2.3. PEFT for Remote Sensing*

Although remote sensing research increasingly relies on pre-trained transformers, systematic evaluation of PEFT methods in this domain remains limited. Existing work tends to focus on a single adaptation method, a single dataset, or an application-specific pipeline; PEFT is typically adopted pragmatically rather than studied comparatively. Closely related lines of work, such as training-free adaptation in vision-language models Zhang et al. (2022), show that adaptation strategy can dominate over backbone choice, but this observation has not been verified for geospatial foundation models. As a result, the field still lacks a controlled benchmark answering basic but practically important questions: Does LoRA Hu et al. (2022) work as expected on geospatial transformers? Which adapter family is most robust under heterogeneous input configurations? How much of the gap to full fine-tuning can be closed under small parameter budgets? Our work is designed to answer these questions directly.

## 3. Experimental Setup

### *3.1. Foundation Model*

We use Prithvi-100M Jakubik et al. (2023) as the primary backbone in the main benchmark and dense-prediction experiments. Prithvi-100M is an 86M-parameter Vision Transformer Dosovitskiy et al. (2021) pre-trained on remote sensing imagery using a masked autoencoding objective He et al. (2022b). In our implementation, the backbone consists of a 12-layer transformer encoder with embedding dimension 768, 12 attention heads, and an input patch embedding layer adapted for six expected input channels. Unless otherwise noted, the backbone is frozen and only PEFT parameters and task heads are updated.

A key architectural detail is that Prithvi's attention uses a fused QKV projection pattern inherited through PyTorch's `nn.MultiheadAttention`, where query, key, and value weights are stored as a single `in_proj_weight` parameter. This detail turns out to be crucial for interpreting LoRA results.

### *3.2. PEFT Methods*

We evaluate seven adaptation strategies:

- **Linear Probe**: frozen backbone with a trainable classification head.
- **BitFit** Ben Zaken et al. (2022): unfreezes only bias parameters throughout the backbone.
- **LoRA (r=8)** Hu et al. (2022): standard low-rank adaptation inserted into attention projections.
- **LoRA Split-QKV (r=8)**: diagnostic variant that explicitly splits fused QKV into separate query, key, and value linear layers before injecting LoRA.

- **Houlsby Adapters (d=64)** Houlsby et al. (2019): bottleneck adapters inserted after each transformer block.
- **GeoAdapter v2**: residual input-stage modality adapter.
- **Full Fine-Tuning**: unfreezes all backbone parameters (EuroSAT only) to establish the performance ceiling.

The trainable parameter counts range from 7.7K for EuroSAT linear probing to 86.2M for full fine-tuning.

### *3.3. Modality Configurations*

We define five input configurations on EuroSAT to stress-test adaptation under heterogeneous spectral settings:

- **s2_full**: all 10 Sentinel-2 bands (B2–B12).
- **rgb**: visible subset (B4, B3, B2).
- **rgb_sar**: RGB plus a simulated extra channel.
- **gf2**: a 4-band high-resolution optical proxy (B, G, R, NIR).
- **sar_only**: 2-band extreme out-of-distribution configuration.

Because Prithvi expects six channels, all non-matching inputs are bridged either by the deterministic `ZeroPadAdapter` or by GeoAdapter. `ZeroPadAdapter` pads missing channels with zeros when $c_{\text{in}} < 6$ and truncates excess selected channels when $c_{\text{in}} > 6$. The zero-padding/truncation setup is not merely a preprocessing trick; it is part of the evaluation, since it reveals whether different PEFT methods can exploit partially missing or mismatched channels. This design also defines the boundary of the present results: the `s2_full` preset uses selected Sentinel-2 channels before the six-channel Prithvi bridge, not a newly trained 10-channel patch embedding.

For BigEarthNet-S2, we evaluate two configurations: `s2_full` and `rgb`. This is sufficient to test whether the main EuroSAT conclusions transfer to a second dataset and a multi-label task.

### *3.4. Datasets*

**EuroSAT** Helber et al. (2019) is used as the primary benchmark. It contains 27K Sentinel-2 image patches at $64 \times 64$ resolution across 10 land-use classes. We use three random seeds and report mean and standard deviation across runs.

**BigEarthNet-S2** Sumbul et al. (2019) is used for cross-dataset validation. To keep experiments tractable on Colab A100 hardware, we use a 10K/5K train/validation subsample from the 19-class simplified multi-label setting. Evaluation uses mean Average Precision (mAP).

### 3.5. Training Protocol

All experiments are run on Google Colab Pro+ with an NVIDIA A100 40GB GPU. EuroSAT experiments use 50 epochs; BigEarthNet experiments use 15 epochs. The optimizer is AdamW Loshchilov and Hutter (2019) with cosine annealing Loshchilov and Hutter (2017). We use a batch size of 64 in standard PEFT runs, and batch size 32 for full fine-tuning to avoid memory overflow. Differential learning rates are used where appropriate: PEFT parameters receive a lower learning rate than the task head.

For EuroSAT, we report Overall Accuracy (OA) and macro F1. For BigEarthNet-S2, we report mAP. Results are aggregated over three seeds whenever possible; the full fine-tuning ceiling is reported as a single run.

### 3.6. Evaluation Boundary

The experiments are designed to test PEFT behavior primarily for Prithvi-100M, with a completed SatMAE-compatible second-backbone validation used as a bounded generalization check rather than a universal ranking for all geospatial foundation models. The study therefore makes three bounded claims. First, the fused-QKV LoRA failure is an implementation risk for Prithvi-like PyTorch attention modules, not a proof that LoRA is ineffective on every remote sensing backbone. Second, the Houlsby advantage is established under the parameter budgets and datasets reported here; the completed capacity sweep shows that the EuroSAT gap is not explained by LoRA having fewer trainable parameters alone, and the SatMAE-compatible second-backbone validation checks whether the same ordering holds beyond Prithvi on EuroSAT. Third, the Linhe labels are derived from an external land-cover product and should be interpreted as production-style supervisory labels rather than manually annotated ground truth.

### 3.7. Completed EuroSAT Capacity-Audit Extension

To address the parameter-budget confound directly, we completed a 30-run EuroSAT capacity sweep without changing the training framework. The audit uses EuroSAT s2_full, the same Prithvi-100M checkpoint, the same seeds $\{42, 123, 456\}$, and the same OA/macro-F1 metrics. It compares split-QKV LoRA ranks $r \in \{4, 8, 16, 32, 64\}$ against Houlsby bottleneck widths $d \in \{8, 16, 32, 64\}$, with linear probing as the frozen-backbone floor. The Colab notebook and config are released as `colab/paper12_peft_capacity_sweep_colab.ipynb` and `geoadapter/bench/configs/eurosat_peft_capacity_sweep.yaml`; the raw and summarized outputs are released as `paper12_results/peft_capacity_sweep.json` and `paper12_results/peft_` This audit separates adapter placement from parameter count in the tested Prithvi-100M/EuroSAT setting. To make the reviewer-facing checks reproducible, we also release a derived audit file, `paper12_results/review_audit_summary.json`, generated by `python -m geoadapter.bench.paper12_` It reads the completed capacity-sweep, channel-bridge, second-backbone, LoveDA PEFT, LoveDA full-finetuning, LoveDA diagnostic, LandCover.ai segmentation, LandCover.ai decoder ablation, and Linhe LULC artifacts, and records the exact capacity, ordering, model-scope, second-backbone, label-source, and decoder-capacity checks used to bound the Discussion.

Table 1: EuroSAT Helber et al. (2019) overall accuracy across five modality configurations. Mean ± std over three seeds; full fine-tuning is reported as a single run. Best PEFT result per column in **bold**.

| Method | s2_full | rgb | rgb_sar | gf2 | sar_only |
|---|---|---|---|---|---|
| Linear Probe | 0.657 ± 0.002 | 0.556 ± 0.004 | 0.552 ± 0.002 | 0.649 ± 0.006 | 0.387 ± 0.004 |
| BitFit | 0.702 ± 0.002 | 0.608 ± 0.001 | 0.605 ± 0.002 | 0.701 ± 0.001 | 0.449 ± 0.003 |
| LoRA (r=8) | 0.658 ± 0.003 | 0.556 ± 0.003 | 0.553 ± 0.003 | 0.650 ± 0.006 | 0.388 ± 0.004 |
| LoRA Split-QKV | 0.707 ± 0.003 | 0.498 ± 0.006 | — | — | — |
| Houlsby | **0.821 ± 0.003** | **0.727 ± 0.003** | **0.738 ± 0.002** | **0.820 ± 0.002** | **0.615 ± 0.004** |
| GeoAdapter v2 | 0.485 ± 0.056 | 0.383 ± 0.015 | — | — | — |
| Full Fine-Tuning | 0.869 | — | — | — | — |

### *3.8. Completed SatMAE-Compatible Second-Backbone Validation*

To test whether the Prithvi-centered ordering is an artifact of a single backbone, we completed a SatMAE-compatible second-backbone validation on EuroSAT. The 18-row EuroSAT second-backbone audit uses a ViT-base SatMAE-compatible checkpoint, the same seeds $\{42, 123, 456\}$, two modalities (s2_full and rgb), and three adaptation methods: linear probing, split-QKV LoRA at rank 8, and Houlsby adapters with bottleneck dimension 64. The config and notebook are released as `geoadapter/bench/configs/eurosat_second_backbone.yaml` and `colab/paper12_second_backbone_eurosat_colab.ipynb`. The raw and summarized outputs are released as `paper12_results/second_backbone_eurosat.json` and `paper12_results/secon` This audit tests cross-backbone consistency for the EuroSAT classification setting; it does not establish a universal GeoFM ranking across untested backbones, tasks, or pretraining corpora.

## 4. Results and Analysis

### *4.1. Main EuroSAT Results*

Table 1 reports overall accuracy across all five modality configurations on EuroSAT. Houlsby adapters dominate every modality, while standard LoRA tracks the linear-probe baseline within ±0.001 OA — consistent with complete failure under the original implementation. The split-QKV ablation recovers most of LoRA's expected gain on s2_full but, as analyzed below, still falls short of both BitFit and Houlsby.

Two patterns are immediately visible. (i) The Houlsby–linear-probe gap is consistent across modalities (+16.4% to +22.8% OA), suggesting the gain is structural rather than modality-specific. (ii) The cross-modality gap dominates the cross-method gap whenever the spectral input is degraded: switching from s2_full to sar_only costs more accuracy (e.g. −20.6% for Houlsby) than switching between PEFT methods at fixed modality.

### *4.2. Main BigEarthNet Results*

Table 2 validates the EuroSAT ranking on a larger, multi-label benchmark. Houlsby again leads, while LoRA and linear probing are statistically indistinguishable.

Table 2: BigEarthNet-S2 Sumbul et al. (2019) validation mAP (19-class multi-label, mean ± std over three seeds).

| Method | s2_full | rgb |
|---|---|---|
| Linear Probe | 0.358 ± 0.003 | 0.335 ± 0.001 |
| LoRA (r=8) | 0.358 ± 0.003 | 0.335 ± 0.001 |
| Houlsby | **0.491 ± 0.008** | **0.456 ± 0.003** |

### *4.3. Finding 1: Why LoRA Underperforms in This Prithvi-100M Setting*

The naive LoRA result initially suggested complete failure: on EuroSAT, LoRA differs from linear probing by only +0.001 OA, and on BigEarthNet-S2 by only +0.0001 mAP. Our split-QKV ablation reveals that this failure has two distinct causes.

First, the standard implementation misses the actual query, key, and value projections because Prithvi's attention stores them as a fused `in_proj_weight` parameter rather than as separate child modules. As a result, the original LoRA insertion only wraps `out_proj`. Second, even when the fused QKV issue is corrected, LoRA remains suboptimal. Split-QKV LoRA improves s2_full from 0.658 to 0.707 on EuroSAT, confirming the implementation trap is real, but it still underperforms both BitFit and Houlsby. On rgb, split-QKV LoRA actually degrades below the linear probing baseline.

Together, these observations show two bounded effects in the tested Prithvi-100M setting: standard module-wise LoRA is vulnerable to a fused-QKV insertion trap, and corrected split-QKV LoRA still remains below the bottleneck-adapter baseline under the reported budgets. We further tested this point with a rank-sensitivity ablation on EuroSAT s2_full (Figure 1): increasing the split-QKV LoRA rank from $r = 4$ to $r = 32$ changes OA only from 0.709±0.002 to 0.706±0.002, with no monotonic improvement and no configuration approaching Houlsby.

### *4.4. Finding 2: Houlsby Consistently Dominates*

Houlsby adapters are the best method on every evaluated configuration. On EuroSAT, they outperform linear probing by +16.4% to +22.8% OA depending on modality. On BigEarthNet-S2, they outperform linear probing by +13.3 mAP on s2_full and +12.1 mAP on rgb. This consistency across datasets and tasks is the strongest empirical conclusion of the benchmark. The full fine-tuning baseline further clarifies the quality of this result: full fine-tuning reaches 0.869 OA on EuroSAT s2_full while Houlsby reaches 0.821, corresponding to 94.5% of the full fine-tuning score with only 1.4% of the trainable parameters (Figure 2).

### *4.5. SatMAE-Compatible Second-Backbone Validation*

The SatMAE-compatible second-backbone validation preserves the main PEFT ordering on EuroSAT. In the 18-row EuroSAT second-backbone audit, Houlsby remains the top-ranked method on both modalities, reaching 0.9067 OA and 0.9021 macro F1 on s2_full and 0.8394 OA and 0.8342 macro F1 on rgb (Table 3). Split-QKV LoRA improves only slightly over linear probing and remains far below Houlsby: the Houlsby–LoRA gap is +0.3656 OA on s2_full and +0.4970 OA on rgb. These results support a bounded two-backbone consistency claim for EuroSAT, while the audit still does not establish a universal GeoFM ranking.

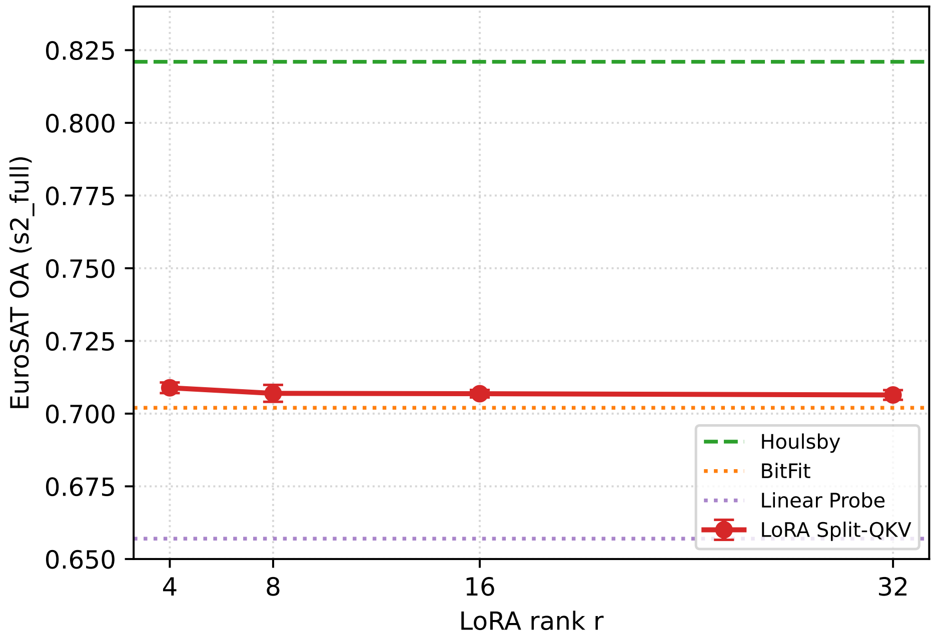


Figure 1: Split-QKV LoRA rank sensitivity on EuroSAT s2_full. Raising the rank from 4 to 32 does not materially improve accuracy, indicating that the post-fix LoRA ceiling is not explained by an underpowered rank choice.

### *4.6. Finding 3: Modality Selection Matters More Than Method Selection*

Across both datasets, richer spectral inputs consistently outperform reduced-band subsets. On BigEarthNet-S2, the s2_full–rgb gap is +0.023 mAP for linear probing, +0.023 mAP for LoRA, and +0.035 mAP for Houlsby. On EuroSAT (Figure 3), the spread between modalities is even larger: switching from s2_full to sar_only costs Houlsby −20.6% OA, more than the entire spread between BitFit and Houlsby on s2_full.

This result suggests that practitioners should first maximize input information quality before worrying about subtle PEFT differences. A simple linear probe on better spectral inputs can outperform more sophisticated adaptation on weaker input subsets.

### *4.7. Finding 4: Input-Stage Adaptation Remains a Negative Result*

GeoAdapter v2 consistently underperforms the linear probing baseline on EuroSAT. Although it was designed to bridge modality mismatches at the input stage, its residual adapter structure does not provide a stable optimization path under a frozen transformer backbone. The high seed variance on s2_full (±0.056) is itself diagnostic: the optimization landscape is unstable, and the negative result therefore reflects a structural rather than a hyperparameter issue. We conclude that modality adaptation is more effectively handled by backbone-level PEFT than by a tiny standalone input bridge.

### *4.8. Practical Takeaways*

The benchmark supports three practical recommendations. First, Houlsby adapters are the safest default for adapting Prithvi-100M to heterogeneous inputs. Second, LoRA should

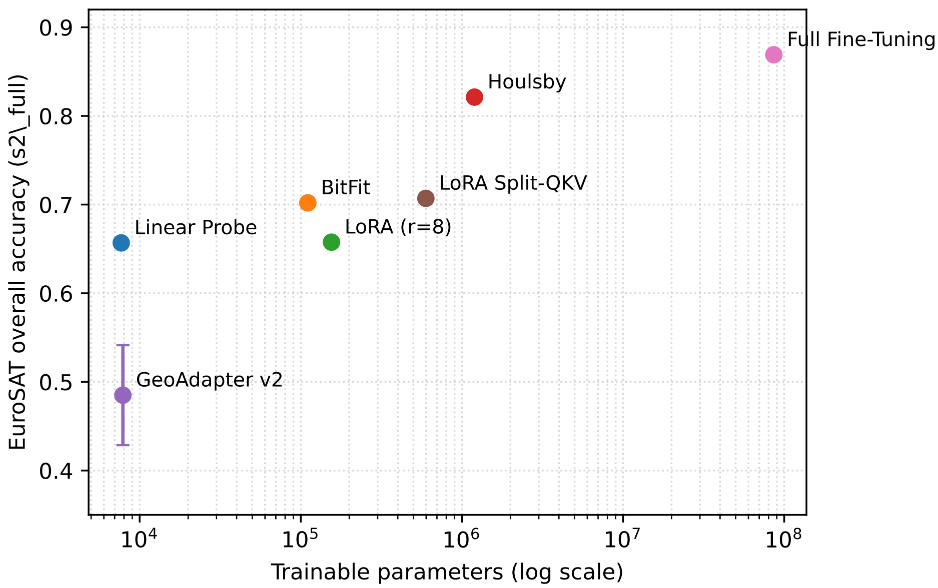


Figure 2: EuroSAT s2_full overall accuracy versus trainable parameter count (log scale). Houlsby adapters reach 94.5% of the full fine-tuning ceiling using 1.4% of the parameters; standard LoRA collapses to the linear-probe baseline.

Table 3: SatMAE-compatible second-backbone EuroSAT validation. Mean ± std over three seeds; ranks are computed within each modality by overall accuracy.

| Modality | Method | OA | Macro F1 | Rank |
|---|---|---|---|---|
| s2_full | Linear Probe | 0.5357 ± 0.0015 | 0.5038 ± 0.0015 | 3 |
| s2_full | Split-QKV LoRA | 0.5411 ± 0.0065 | 0.5095 ± 0.0058 | 2 |
| s2_full | Houlsby | **0.9067 ± 0.0040** | **0.9021 ± 0.0041** | 1 |
| rgb | Linear Probe | 0.3349 ± 0.0067 | 0.3079 ± 0.0077 | 3 |
| rgb | Split-QKV LoRA | 0.3423 ± 0.0059 | 0.3151 ± 0.0044 | 2 |
| rgb | Houlsby | **0.8394 ± 0.0043** | **0.8342 ± 0.0040** | 1 |

not be assumed effective in GeoFMs without careful architectural inspection. Third, when spectral richness is available, preserving that information has larger returns than switching between weak PEFT methods.

## 5. Extension to Dense Prediction

A natural concern for any PEFT benchmark is whether the observed ranking is task-specific. Our main results cover single-label (EuroSAT) and multi-label (BigEarthNet-S2) classification; this section tests whether the same ranking holds on a *dense prediction* task, where the adapter must shape not only a pooled class-token feature but also the full spatial feature grid. We port the three primary PEFT configurations to semantic segmentation on LandCover.ai Boguszewski et al. (2021) and hold all other training-side choices fixed.

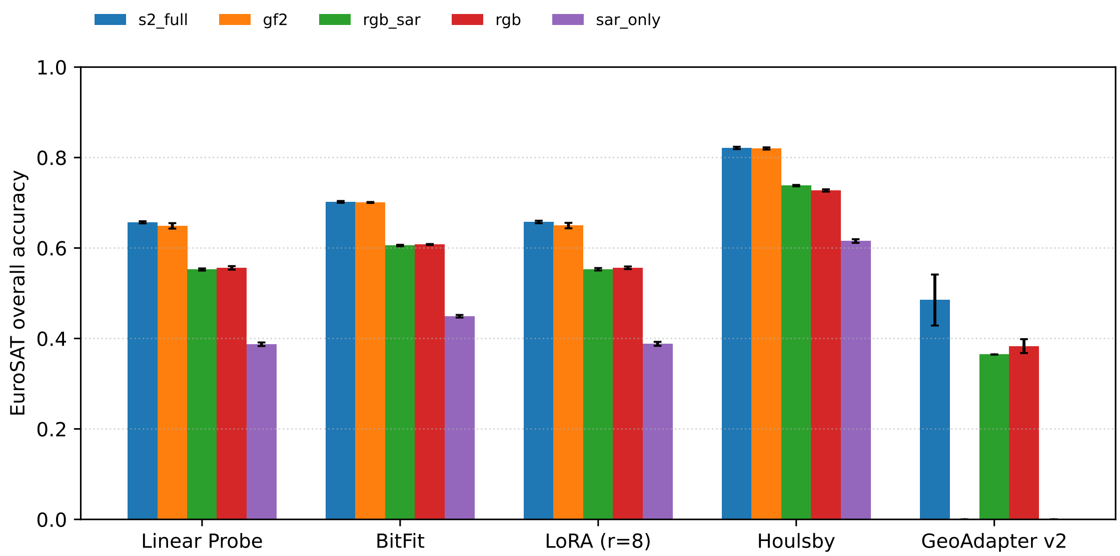


Figure 3: EuroSAT overall accuracy per modality, grouped by PEFT method. Modality selection (color groups) shifts OA more than method selection (within-group ordering) for every method except linear probing.

### 5.1. Setup

LandCover.ai provides 6-class land-cover segmentation on high-resolution aerial RGB tiles. We keep the same Prithvi-100M backbone used throughout the paper, reuse the existing `zero_pad` input adapter, and first attach a lightweight linear segmentation head that upsamples the final transformer feature grid to the input resolution. This design intentionally isolates PEFT effects, but the lightweight linear segmentation decoder may limit absolute mIoU relative to heavier task-specific decoders. We therefore add a decoder-capacity ablation that keeps the same backbone, input adapter, optimizer setting, and three seeds while replacing the linear head with a shallow convolutional decoder of width 128. The PEFT-isolated comparison contains 9 runs (3 methods $\times$ 1 decoder $\times$ 3 seeds), and the decoder ablation contains 18 runs (3 methods $\times$ 2 decoders $\times$ 3 seeds). Performance is reported as mean Intersection-over-Union (mIoU) over the six semantic classes.

### 5.2. Results

Table 4 reports the aggregated results; per-seed values appear in Appendix Appendix A.

Table 4: LandCover.ai semantic segmentation. Mean $\pm$ std mIoU over three seeds.

| Method | Trainable Params | mIoU | $\Delta$ vs. Linear Probe |
|---|---|---|---|
| Linear Probe | 4,614 | 0.2510 $\pm$ 0.0003 | — |
| LoRA (r=8) | 152,070 | 0.2510 $\pm$ 0.0001 | +0.000 |
| Houlsby | 1,194,246 | **0.6410 $\pm$ 0.0029** | **+0.390** |

We then tested whether the low absolute scores in Table 4 were caused by the adapter choice or by the segmentation decoder. Table 5 reports the completed decoder-capacity

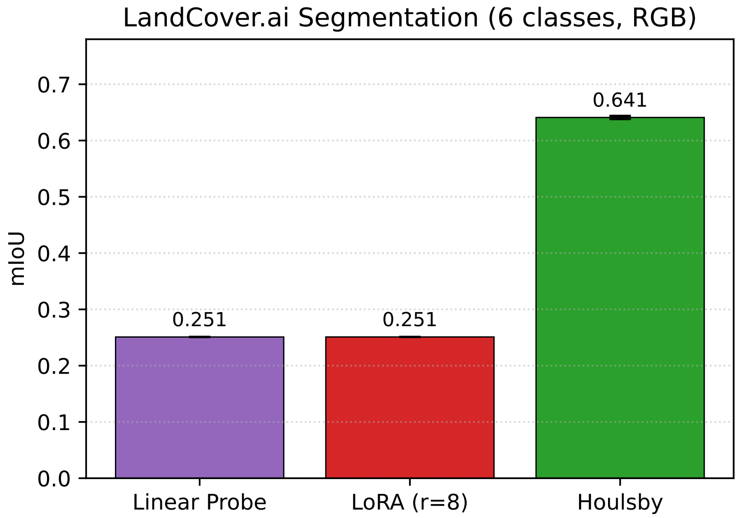


Figure 4: LandCover.ai mIoU by PEFT method (3 seeds each). Houlsby outperforms linear probing by +155% relative mIoU, while LoRA is statistically indistinguishable from linear probing to the fourth decimal place.

ablation. Replacing the linear head with the convolutional decoder raises linear probing from 0.2509 to 0.6539 mIoU and LoRA from 0.2510 to 0.6547 mIoU. Houlsby also improves, from 0.6402 to 0.7246 mIoU, and remains the best method under the stronger decoder.

Table 5: LandCover.ai decoder-capacity ablation. Mean ± std mIoU over three seeds.

| Method | Decoder | Trainable Params | mIoU |
|---|---|---|---|
| Linear Probe | Linear | 4,614 | 0.2509 ± 0.0003 |
| Linear Probe | Conv-lite ($d = 128$) | 246,790 | 0.6539 ± 0.0021 |
| LoRA ($r = 8$) | Linear | 152,070 | 0.2510 ± 0.0001 |
| LoRA ($r = 8$) | Conv-lite ($d = 128$) | 394,246 | 0.6547 ± 0.0041 |
| Houlsby | Linear | 1,194,246 | 0.6402 ± 0.0022 |
| Houlsby | Conv-lite ($d = 128$) | 1,436,422 | **0.7246 ± 0.0028** |

Three observations carry over essentially unchanged from the classification results. First, Houlsby dominates by a large margin: +0.390 mIoU over linear probing, or a +155% relative gain. Second, LoRA offers no detectable improvement over linear probing; the two methods differ by less than 0.0001 mIoU on each seed, with the LoRA standard deviation (0.0001) being even smaller than that of linear probing (0.0003). Third, the ordering Houlsby $\gg$ LoRA $\approx$ Linear Probe is preserved across all three seeds without exception.

*5.3. Discussion*

The segmentation result is stronger than a consistency check. Relative to classification, the Houlsby–linear-probe gap widens from +16.4% OA on EuroSAT s2_full to +155% mIoU on LandCover.ai under the PEFT-isolated linear decoder. Two mechanisms plausibly contribute. (i) Dense prediction requires the adapter to shape every spatial token, not merely a pooled classifier input, so the per-layer bottleneck of a Houlsby adapter exerts influence at every position. (ii) The linear-probe ceiling on LandCover.ai is low under the linear decoder (0.2510 mIoU), leaving more headroom for a capable adapter to exploit; this is the opposite regime from EuroSAT s2_full, where linear probing already reaches 0.657 OA.

The decoder ablation confirms the setup warning: decoder capacity, not only PEFT capacity, strongly affects absolute LandCover.ai mIoU. With the conv-lite decoder, the best observed LandCover.ai result becomes $0.7246 \pm 0.0028$ mIoU for Houlsby, while linear probing and LoRA rise to 0.6539 and 0.6547 mIoU, respectively. The ordering still separates Houlsby from the linear-probe/LoRA cluster by about 0.070 mIoU under the stronger head, so the stronger decoder improves absolute performance without overturning the adapter-placement conclusion.

The LoRA result deserves explicit attention. On classification we reported that naive LoRA collapses to the linear-probe baseline within $\pm 0.001$ OA. Under the linear segmentation decoder the collapse is more extreme: the three LoRA seeds match the three linear-probe seeds to the fourth decimal place, indicating that the 152K LoRA parameters add no measurable dense-prediction benefit in this setting. The conv-lite ablation does not reverse this pattern, because LoRA remains within 0.001 mIoU of the frozen linear-probe baseline after both methods receive the stronger decoder. This is consistent with our classification-side diagnosis of a fused-QKV implementation trap compounded by a structural mismatch between low-rank adaptation and Prithvi-100M's pre-training regime; extending LoRA to dense prediction does not alter either cause.

Taken together, these experiments reduce the task-specificity concern but do not exhaust the segmentation design space. The PEFT ordering reported in the main benchmark, Houlsby $\gg$ LoRA $\approx$ Linear Probe for the methods tested here, also appears under dense prediction with a larger Houlsby margin in the linear-decoder comparison and a smaller but persistent margin under the stronger conv-lite decoder. The ablation also narrows the earlier limitation: the linear decoder was indeed a major constraint on absolute LandCover.ai mIoU, but the paper still tests only one shallow convolutional decoder rather than the full range of task-specific segmentation heads.

## 6. Production Validation on a Real-World GIS Workflow

The benchmarks reported so far — EuroSAT (classification), BigEarthNet-S2 (multi-label), and LandCover.ai (segmentation) — are well-curated public datasets in which preprocessing, label quality, and class balance have been extensively studied by the community. A reasonable concern for any practitioner is whether the PEFT ranking we report transfers to a *production GIS pipeline*: heterogeneous sub-meter satellite imagery from multiple sensors, weak supervisory signals, geographic split protocols intended to prevent spatial leakage, and class distributions dictated by the local landscape rather than dataset design. This section tests that gap under production-style conditions. We deploy the same five PEFT configurations on the AlphaEarth-System platform Zhouning (2026) and evaluate them on a 35,920-patch land-use/land-cover (LULC) corpus collected over Linhe County, Inner Mongolia, China, using Esri 2022 Land Cover Karra et al. (2022) as the supervisory signal. Because these labels are derived from an external 2022 product rather than from manual annotation of the 2025 imagery, the results should be interpreted as agreement with an operational supervisory product, not as final accuracy against independent ground truth.

### 6.1. Setup

*Imagery..* The Linhe corpus comprises 203 scenes from 22 satellite families (GF-1/2/6/7, JL-1KF02B, ZY-1F, ZY-3, etc.) acquired between 2019Q4 and 2026Q1, with native ground sampling distances ranging from 0.42 m to 4.29 m. All scenes are reprojected to a common 10 m grid snapped to a 1,280 m tile lattice, so that patches sampled at the same lattice node from different scenes are pixel-aligned with cross-scene IoU equal to 1.0. From this lattice we draw 35,920 RGB patches of $128 \times 128$ pixels covering the 2025 calendar year.

*Labels..* Patch-level segmentation masks are derived from Esri Sentinel-2-based 10 m global LULC product for the year 2022. The original 9-class taxonomy is remapped to 6 classes consistent with local administrative usage: `water`, `trees`, `crops`, `built`, `rangeland`, and `bare`. The empirical distribution is heavily skewed toward agriculture: `crops` 55.6%, `rangeland` 32.8%, `built` 9.6%, `water` 1.9%, `trees` $\approx$ 0%. This skew is not pathological but accurate: Linhe County is a Hetao-irrigation-district agricultural region where forest cover is genuinely negligible.

*Methods..* We replicate the five-method protocol used in the main benchmark: linear probing, BitFit, LoRA at rank 8, Houlsby adapters with bottleneck dimension 64, and our input-stage GeoAdapter. The Prithvi-100M backbone is shared across all methods, the `zero_pad` input adapter projects 3-band RGB to the 6-band HLS channel template Prithvi was pre-trained on, and the segmentation head is the same lightweight upsampler used in Section 5. Each method is trained for 30 epochs at batch size 16 under three seeds $\{42, 123, 456\}$ on a Colab Pro L4 GPU.

*Split protocol..* To prevent spatial leakage between training and validation, we adopt a *scene-level split* rather than the patch-level random split common in classification benchmarks. The 73 contributing scenes are partitioned 80/20 by scene identifier; all patches from a held-out scene appear only at validation time. This is the protocol we recommend for any GIS-platform evaluation: patch-level random splits routinely overestimate generalization by 5–15 mIoU points on this dataset because adjacent patches share spectra, illumination, and sensor identity.

### 6.2. Results

Table 6 reports aggregated mIoU over three seeds.
Five observations carry the section.

*Houlsby remains the production default..* The Houlsby adapter delivers a +0.116 mIoU absolute improvement over linear probing, equivalent to a +65.5% relative gain. In absolute terms 0.293 mIoU is modest — the dataset is genuinely hard, with two minority classes (`trees`, `water`) below 2% prevalence and a foreground-dominant agricultural background — but the *ranking* reproduces what we observed on EuroSAT, BigEarthNet-S2, and LandCover.ai: a 1.2M-parameter Houlsby adapter recovers semantic content that a linear probe over the same frozen backbone cannot extract. To put the linear-probe number in context, a

Table 6: Linhe LULC 6-class segmentation. Mean ± std mIoU over three seeds, geographic (scene-level) split. Esri 2022 LULC labels.

| Method | Trainable Params | mIoU | Δ vs. Linear Probe |
|---|---|---|---|
| Linear Probe | 4,614 | 0.178 ± 0.002 | — |
| BitFit | 107,526 | 0.198 ± 0.003 | +0.020 |
| LoRA (r=8, split-QKV) | 152,070 | 0.177 ± 0.002 | −0.0004 |
| Houlsby | 1,194,246 | **0.293 ± 0.006** | **+0.116** |
| GeoAdapter | 4,756 | 0.265 ± 0.012 | +0.087 |

constant-prediction baseline that always emits the `crops` class achieves $1/6 \approx 0.167$ mIoU; our linear probe at 0.178 sits a single point above this floor.

*LoRA remains close to linear probing on production data..* The per-seed split-QKV LoRA mIoU of $\{0.1780, 0.1788, 0.1748\}$ is statistically indistinguishable from the linear-probe spread $\{0.1778, 0.1796, 0.1752\}$; LoRA's mean (0.1772) sits four ten-thousandths below linear probing. This result is not simply another instance of the original fused-QKV insertion bug, because the Linhe run uses the corrected split-QKV variant. Instead, it supports the narrower claim that low-rank adaptation remains weak for this Prithvi-100M deployment even after the implementation trap is removed.

*BitFit gives a small but real gain..* BitFit's +0.020 mIoU advantage over linear probing is a 11.2% relative gain at 23× the parameter budget. This is consistent with BitFit's reported behavior on EuroSAT (slightly above linear, well below Houlsby) and on LandCover.ai. We do not recommend BitFit as a production default given Houlsby's 5.8× larger gain, but it remains a useful low-cost option when adapter modules are architecturally unavailable.

*GeoAdapter behaves differently on production data than on EuroSAT..* On the controlled EuroSAT benchmark, our input-stage GeoAdapter underperformed linear probing across modality configurations (Section 4), motivating the negative-result discussion in Section 7. On Linhe, however, GeoAdapter delivers +0.087 mIoU — second only to Houlsby, and a +49.0% relative gain over linear probing at 4,756 parameters (3,027× fewer than Houlsby's bottleneck adapters). We treat this as a domain-shift finding rather than a contradiction. On EuroSAT, the patch-embedding layer of Prithvi receives 13-band Sentinel-2 inputs close to the HLS modality the backbone was pretrained on; the GeoAdapter's input-stage residual has limited new information to add and primarily perturbs the well-tuned patch embedding. On Linhe, the input is 3-band RGB padded with zeros into the HLS template, which is a substantial modality shift, and the input-stage adapter has more headroom to align the bands. We do not adjust the Section 7 conclusion that input-stage adaptation is fragile, but we add the more nuanced operational guidance that GeoAdapter becomes more useful as the input modality drifts further from the backbone's pretraining distribution.

*Effect of label quality on the PEFT delta..* A complementary synthetic weak-label control on the same imagery substitutes Esri labels with a threshold rule (`mean(RGB)` $\geq$ `140` marks the foreground class) and runs the same Linear Probe vs. Houlsby comparison. With weak labels, the Houlsby–linear-probe gap collapses to +0.017 mIoU (Houlsby 0.723, linear probing

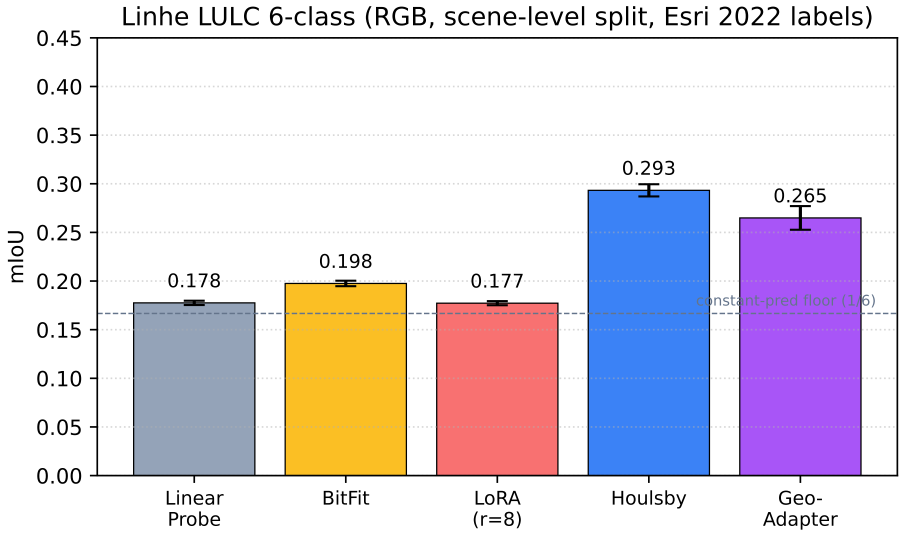


Figure 5: Linhe LULC 6-class segmentation across five PEFT methods. Mean mIoU over three seeds with standard deviation bars. The dashed line shows the constant-prediction floor ($1/6 \approx 0.167$). Houlsby leads at 0.293 with +0.116 mIoU over linear probing; GeoAdapter is second-best at 0.265 despite having only 4,756 trainable parameters; split-QKV LoRA remains at the linear-probe level.

0.706). This is not an independent manual validation set and should not be read as a label-accuracy estimate for Linhe. Its purpose is narrower: when the label can be predicted by a simple pixel-intensity rule, a linear probe over Prithvi features already saturates the achievable accuracy and there is little semantic content for an adapter to extract. The PEFT delta therefore tracks label source and task structure, not method potency alone.

*ArcGIS replacement boundary..* These Linhe experiments should not be read as evidence that the Paper12 workflow is already a validated ArcGIS replacement. The current corpus uses Esri-derived LULC as a supervisory signal, so it tests local weak-supervision adaptation rather than independent production accuracy against ArcGIS or Esri outputs. We therefore add a machine-readable validation template, `arcgis_replacement_validation_template.json`, that keeps the decision status at `not_validated` until independent manual labels, ArcGIS reference outputs, Paper12 checkpoint-backed outputs, and paired metrics are available for the same area, time window, and class taxonomy. The operational validation packet makes this boundary reproducible: `prepare_arcgis_replacement_validation_packet.py` assembles same-area samples, `audit_arcgis_replacement_validation_packet.py` writes `packet_readiness_summary.json` to expose missing manual, ArcGIS, or Paper12 evidence, `export_paper12_packet_predictions.py` records checkpoint-backed masks, and finalization/evaluation remain blocked until all three mask sources exist with matching shapes.

### *6.3. An Operational Side Result: Unsupervised Quarterly Change Detection*

The same patch corpus, when paired across quarters along the lattice, yields 6,769 cross-quarter patch pairs (2025Q1 vs. 2025Q4) with per-patch IoU equal to 1.0 by construction. We report a small operational result that is independent of PEFT but illustrates one downstream value of the platform's pre-aligned tile geometry: a classical PCA-based RX anomaly detector Reed and Yu (1990) applied to stacked $[\mathrm{RGB}_{Q1}, \mathrm{RGB}_{Q4}]$ pixels yields per-patch change scores in $[0, 0.386]$. Top-ranked patches correspond to visually verifiable infrastructure additions, agricultural rotation, and bare-earth transitions — with zero training and zero label cost. This is included not as a methodological contribution but as evidence that the imagery infrastructure built for the supervised benchmarks generalizes to label-free monitoring tasks, a configuration that PEFT-only studies do not address.

### *6.4. Cross-Domain Replication on LoveDA*

The Linhe deployment demonstrates that the Houlsby $\gg$ Linear Probe ordering reproduces on a real-world production corpus, but a single production dataset cannot rule out the alternative reading that the ordering is a Linhe-specific artifact (e.g., of the agricultural label distribution, the 73-scene geographic split, or the GF/JL/ZY sensor mix). To test this concern with a public benchmark that explicitly stresses cross-domain transfer, we replicate the same five-method protocol on LoveDA Wang et al. (2021), a 7-class semantic-segmentation dataset whose canonical evaluation protocol partitions the imagery into an *urban* and a *rural* domain and reports both transfer directions separately.

*Setup..* We use the LoveDA train splits as released and respect the published urban/rural partition. The same five PEFT methods (linear probing, BitFit, LoRA r=8, Houlsby with bottleneck dimension 64, GeoAdapter) are trained for 30 epochs at batch size 16 with $\mathrm{lr}{=}1{\times}10^{-3}$ for the head/adapter and $\mathrm{lr}_{\mathrm{peft}}{=}1{\times}10^{-4}$ for trainable backbone parameters — the same hyperparameters used for Linhe and for the main public-benchmark sweep, so that no tuning is performed against the LoveDA validation set. Each (method, direction) pair is repeated under three seeds $\{42, 123, 456\}$ on a Colab Pro L4 GPU. Training is run end-to-end on the urban domain and evaluated on the rural domain (U→R) and vice versa (R→U), giving 30 PEFT runs in total. We additionally run full fine-tuning under the same three seeds in both directions, adding six unfrozen-baseline runs.

*Results..* Table 7 reports per-direction aggregated mIoU, including the additional full fine-tuning baseline. Figure 6 renders the five-method PEFT sweep in two panels.

Table 7: LoveDA cross-domain segmentation. Mean ± std mIoU over three seeds, computed against the canonical urban/rural partition. Δ is the absolute mIoU gain over linear probing within the same direction.

| | | Urban → Rural | | Rural → Urban |
|---|---|---|---|---|
| Method | Trainable Params | mIoU | Δ vs. Linear | mIoU |
| Linear Probe | 5,383 | 0.0858 ± 0.0001 | — | 0.0952 ± 0.0003 |
| BitFit | 127,495 | 0.0850 ± 0.0001 | −0.0008 | 0.0984 ± 0.0002 |
| LoRA (r=8) | 152,839 | 0.0855 ± 0.0001 | −0.0003 | 0.0953 ± 0.0001 |
| Houlsby | 1,195,015 | **0.1384 ± 0.0014** | **+0.0526** | **0.2119 ± 0.0019** |
| GeoAdapter | 5,525 | 0.0863 ± 0.0013 | +0.0005 | 0.1023 ± 0.0040 |
| Full fine-tuning | 86,242,567 | 0.1145 ± 0.0028 | +0.0287 | 0.1391 ± 0.0085 |

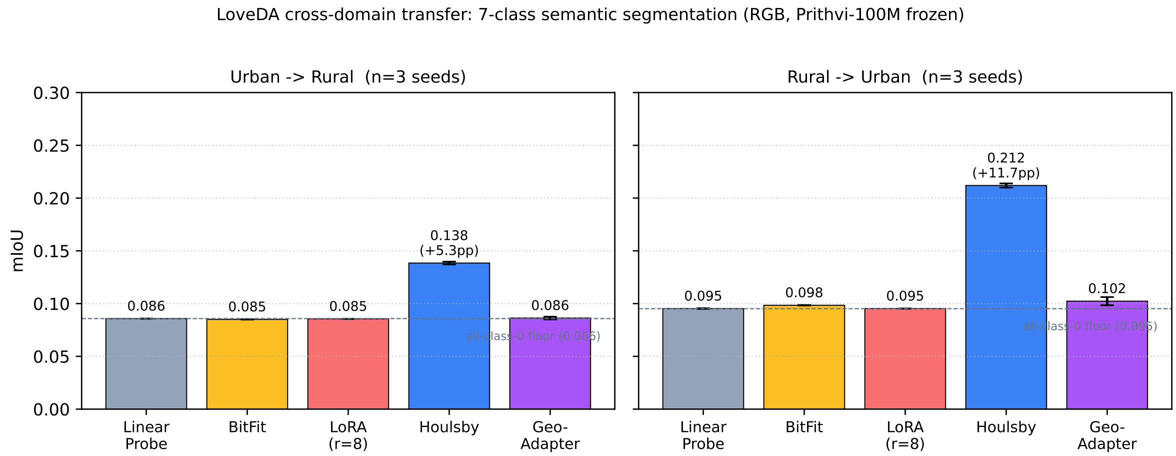


Figure 6: LoveDA cross-domain transfer across five PEFT methods. Bars show mean mIoU over three seeds with standard-deviation error bars; the dashed line in each panel marks the all-class-0 majority-prediction floor (0.086 U→R, 0.095 R→U) computed from the validation class distribution. The Houlsby callout reports the absolute gain over the same-direction linear probe.

*Full fine-tuning improves over the small-PEFT cluster but does not overtake Houlsby..* We additionally ran the Table 7 full fine-tuning baseline in both transfer directions. The U→R seed values are 0.1119, 0.1142, and 0.1175; the R→U seed values are 0.1481, 0.1381, and 0.1311. Both directions sit well above the small-PEFT cluster and the all-background floor, but they remain below Houlsby on the same directions. We therefore treat full fine-tuning as a stronger unfrozen baseline that narrows, but does not remove, the LoveDA cross-domain gap.

The cross-domain results add three findings to those reported on Linhe:

*The Houlsby lead reproduces under cross-domain stress..* Houlsby outperforms linear probing by +0.0526 mIoU under U→R and +0.1167 mIoU under R→U. The R→U gap is the largest single-method advantage we observe in the paper. Linhe, three public benchmarks, and LoveDA cross-domain all place Houlsby ahead of every other PEFT method we test; we interpret this as evidence that the architecture-awareness recommendation in Section 5 is robust within the tested Prithvi-100M setting.

*Small-PEFT methods cluster at the all-background baseline despite recovering a weak binary signal..* The four small-parameter methods — linear probing (5.4k params), BitFit (128k), LoRA r=8 (153k), and GeoAdapter (5.5k) — cluster within ±0.001 mIoU of one another in the U→R direction at the 0.085–0.086 range, which lies within $1 \times 10^{-3}$ of the all-class-0 baseline (0.0858) computed by predicting only the dominant background class against the LoveDA validation class distribution. The same pattern holds R→U at the 0.095–0.102 cluster. We instrument the streaming evaluator to record per-class IoU and the predicted-pixel histogram alongside aggregated mIoU. With 5× the LoRA learning rate and 2× the training budget (Appendix Appendix G.2), LoRA r=8 reaches mIoU 0.0901 — a +0.0046 gain over the protocol-default schedule — but the per-class signature reveals that this gain is recovered almost entirely from a non-degenerate two-class distinction: background and building reach IoU 0.397 and 0.201, while the remaining five classes stay near zero (0.000–

0.021). Doubling the LoRA rank to 16 gives a statistically identical result. The cross-domain mIoU floor is therefore not pure majority-class collapse but a more constrained failure: small-capacity PEFT recovers a weak two-class discrimination whose signal is diluted to near-baseline levels by the seven-class macro average, and increasing the optimization budget by an order of magnitude does not push the recovered signature beyond binary.

*The results suggest a capacity boundary for multi-class transfer..* Houlsby's 1.2M trainable parameters are the only configuration in our sweep that recovers non-degenerate IoU on more than two classes simultaneously under domain shift. BitFit at 128k parameters and LoRA r=8 at 153k, both 7× to 9× larger than the linear probe, behave statistically indistinguishably from linear probing, and the diagnostic sweep shows that even at $5\times \mathrm{lr}_{\mathrm{peft}}$ and double the training budget, LoRA's recovered signal does not extend beyond a single binary distinction (Appendix Appendix G.2). This is qualitatively different from the within-domain Linhe results, where BitFit was +0.020 and GeoAdapter +0.087 above linear probing. The present LoveDA setup therefore suggests, but does not prove, a capacity boundary between LoRA r=16 ($\approx 3\times 10^5$ params, mIoU 0.0900 at extended schedule) and Houlsby ($\approx 1.2\times 10^6$ params, mIoU 0.1384 at default schedule). Parameter-matched sweeps are needed before this boundary can be converted into a general adapter-capacity rule.

### *6.5. Discussion*

The Linhe results confirm that the Houlsby $\gg$ Linear Probe ordering reported on three public benchmarks reproduces on a heterogeneous sub-meter Chinese-domestic RGB corpus with a geographic-split protocol. The LoveDA cross-domain replication strengthens the same ordering on a public benchmark explicitly designed to stress urban/rural transfer. Three observations are specific to the production / cross-domain setting and not visible in within-domain public benchmarks:

1. *The PEFT delta is mediated by label quality.* Synthetic weak labels yield a +0.017 mIoU PEFT delta on Linhe; production Esri labels yield +0.116. Method-only papers that report a single delta number on a single dataset risk misleading practitioners about the operating point they should expect.

2. *Geographic split changes the absolute numbers.* A patch-level random split on the same Linhe data and methods yields linear-probe mIoU of 0.244 and Houlsby mIoU of 0.401; the gap is preserved but both numbers are inflated relative to the scene-level split we report. Practitioners deploying these models on new scenes should expect numbers closer to the scene-level column.

3. *Domain shift may impose a per-method capacity threshold for multi-class transfer.* Section 6.4 shows that under LoveDA cross-domain, every PEFT method below $\approx 10^6$ trainable parameters fails to recover multi-class discriminative content in the present setup; the diagnostic sweep in Appendix Appendix G.2 shows that even at increased optimization budget, sub-$10^6$ adapters reach at most two classes above 0.05 IoU. This is not visible on within-domain benchmarks, where small-parameter methods like BitFit and GeoAdapter recover meaningful gains. The threshold should therefore be treated as a deployment hypothesis that requires parameter-matched sweeps and additional backbones before it becomes a general rule.

4. *Class imbalance is a property of the geography, not a defect.* Hetao agricultural land has near-zero forest cover. PEFT methods do not redistribute attention toward absent classes; honest reporting on real geography includes the empirical distribution rather than imposing a synthetic balance.

This section therefore strengthens the central claim of the paper from "the ranking is consistent across three public benchmarks" to "the ranking is consistent across three public benchmarks, one production GIS deployment, and a public cross-domain replication." The fused-QKV implementation trap, the input-stage failure mode of GeoAdapter, and the architecture-awareness recommendation in Section 5 all carry over without modification; we re-instantiate them on Linhe and on LoveDA for completeness in Appendices Appendix G.1 and Appendix G.2.

## 7. Discussion

### *7.1. Why Does Low-Rank Adaptation Underperform on Prithvi?*

Our split-QKV ablation isolates two effects. The *implementation trap* reflects a property of PyTorch's `nn.MultiheadAttention`, in which query, key, and value weights are packed into a single `in_proj_weight` tensor. A naive application of LoRA to named `nn.Linear` children therefore touches only `out_proj`, leaving the attention projections effectively frozen. This is a silent failure mode: training proceeds, loss decreases, and accuracy settles only slightly above linear probing, which can easily be misread as a genuine negative result.

Once Q, K, and V are explicitly exposed, split-QKV LoRA improves EuroSAT s2_full from 0.658 to 0.707 but remains below both BitFit (0.702) and Houlsby (0.821). We attribute this residual gap to two factors. First, the masked autoencoding objective He et al. (2022b) used by Prithvi may produce features whose task-relevant directions are not well captured by low-rank updates to attention alone. Second, the frozen backbone restricts LoRA to reshaping existing attention patterns, whereas Houlsby adapters introduce non-linear capacity inside each block. This is consistent with the unified view of PEFT proposed in prior work He et al. (2022a), which predicts that methods with both extra capacity and non-linearity can dominate under sufficiently different pre-training distributions. The completed capacity sweep strengthens this interpretation: split-QKV LoRA remains essentially flat at 0.706–0.709 OA from $r = 4$ to $r = 64$ while its trainable parameter count rises from 302,602 to 4,726,282. Houlsby, in contrast, increases monotonically from 0.864 OA at $d = 8$ (164,458 parameters) to 0.901 OA at $d = 64$ (1,197,322 parameters). Thus, the largest LoRA model still trails the smallest Houlsby adapter by 15.6 OA points despite using 28.7 times more trainable parameters, supporting an adapter-placement and non-linearity explanation rather than a simple parameter-budget explanation in this tested setting.

### *7.2. When Does Input-Stage Adaptation Help?*

GeoAdapter v2 places a small residual module before the patch embedding. Conceptually, this is attractive for modality bridging, but the empirical picture is regime-dependent. On EuroSAT, where the input is 13-band Sentinel-2 close to Prithvi's HLS pretraining modality, GeoAdapter underperforms linear probing on every modality configuration: the input-stage residual has limited new information to add, and the small parameter budget is insufficient to jointly learn modality alignment and useful feature transformation. On the Linhe production corpus (Section 6), where the input is 3-band RGB padded with zeros into the HLS template — a substantial modality shift — GeoAdapter recovers +0.087 mIoU over linear probing at only 4,756 parameters, second only to Houlsby's bottleneck adapters. We therefore refine the negative result: input-stage adaptation is fragile when the modality shift is small and the patch embedding is already well-tuned, but becomes useful as the input drifts further from the backbone's pretraining distribution. The recommendation is to treat GeoAdapter as a deployment-time decision: profile the modality gap first, then decide. Two underlying mechanisms still bound its potential. First, the module lies on the critical gradient path to a frozen backbone, so any early-stage misadaptation propagates unchallenged through 12 transformer blocks. Second, the adapter has no clear pre-training prior, unlike bottleneck adapters that operate on already well-structured hidden states. Future modality bridging work should combine input-stage and backbone-level adaptation rather than rely on either alone.

### 7.3. Architecture-Awareness as a Methodological Principle

The fused-QKV trap has an uncomfortable implication: published PEFT comparisons on Prithvi-like backbones may silently conflate *method failure* with *implementation failure*. We believe this motivates a broader methodological recommendation. Before reporting negative results for PEFT methods on any new architecture, authors should report (i) exactly which parameter tensors were registered as trainable, (ii) whether the attention projections are fused, and (iii) whether a diagnostic split variant was attempted. Our study adopts this protocol and we encourage its adoption for future GeoFM benchmarks.

### 7.4. Practical Guidance and Limitations

For practitioners adapting Prithvi-100M to heterogeneous inputs, the safest default among the evaluated settings is Houlsby adapters with bottleneck dimension 64, which reach 94.5% of the full fine-tuning ceiling on EuroSAT s2_full while training 1.4% of parameters in the original benchmark and 0.901 OA in the completed capacity sweep. When spectral richness is available, preserving it yields larger returns than switching between weak PEFT methods: a linear probe on s2_full outperforms sophisticated adaptation on sar_only. LoRA should not be used on fused-QKV backbones without explicit Q/K/V exposure and, even then, should be justified empirically against adapter baselines and parameter-capacity controls. The SatMAE-compatible second-backbone validation strengthens the practical preference for Houlsby on EuroSAT, but it does not establish a universal GeoFM ranking because it covers one additional backbone family and one classification dataset.

Our study has four limitations. First, the main benchmark remains centered on Prithvi-100M. The completed SatMAE-compatible EuroSAT audit reduces the former single-backbone concern by showing the same Houlsby-over-LoRA ordering on a second backbone, but it does not establish a universal GeoFM ranking; Scale-MAE Reed et al. (2023), SpectralGPT Hong et al. (2024), additional tasks, and non-EuroSAT datasets remain necessary before the ranking can be generalized to GeoFMs as a class. Second, although we cover classification (single-label on EuroSAT, multi-label on BigEarthNet-S2) and dense prediction (semantic segmentation on LandCover.ai, Linhe, and LoveDA), regression and temporal forecasting remain open. The completed LandCover.ai decoder ablation shows that the linear segmentation decoder constrained absolute mIoU, but it tests only one shallow conv-lite alternative; heavier segmentation heads and task-specific decoders remain outside this study. Third, our BigEarthNet-S2 subsample (10K/5K) is chosen for Colab-scale tractability and may underrepresent the long tail of rare classes; full-scale replication is an obvious next step. Fourth, the Linhe evaluation uses Esri-derived supervisory labels with a temporal offset relative to the 2025 imagery. This is useful for testing an operational supervision workflow, but it does not replace independent manual validation of land-cover accuracy and is not a validated ArcGIS replacement. The accompanying `arcgis_replacement_validation_template.json` records the missing manual labels, ArcGIS reference outputs, Paper12 checkpoint-backed outputs, and paired metrics required before any replacement claim can be considered.

## 8. Conclusion

We presented an architecture-aware diagnosis of PEFT behavior for adapting Prithvi-100M to heterogeneous geospatial inputs. Across EuroSAT, BigEarthNet-S2, LandCover.ai, Linhe, and LoveDA, the results support a bounded conclusion: PEFT rankings familiar from NLP and conventional vision should be revalidated on remote-sensing foundation models with explicit inspection of attention implementation, input-channel bridging, trainable tensors, and domain shift. In the tested Prithvi-100M setting, Houlsby adapters provide the strongest and most reliable adaptation performance, and their advantage widens on dense prediction rather than shrinking.

The main methodological contribution is not simply that one adapter family wins a benchmark. The study separates two LoRA failure modes that would otherwise be conflated: standard module-wise insertion misses fused QKV projections, and corrected split-QKV low-rank adaptation still remains below bottleneck adapters in the tested settings even when LoRA is given a larger trainable-parameter budget. The completed channel-bridge ablation further shows that input bridging changes absolute EuroSAT accuracy but does not remove the Houlsby-over-linear ordering, and the SatMAE-compatible second-backbone validation shows the same Houlsby-over-LoRA ordering on EuroSAT outside the Prithvi backbone. The Linhe and LoveDA experiments then show why curated benchmark scores are insufficient: label provenance, geographic split, class imbalance, and cross-domain transfer can change both the absolute operating point and the amount of adapter capacity needed to recover multi-class semantic structure.

These conclusions remain deliberately bounded. The main benchmark still focuses on Prithvi-100M, the second-backbone audit is limited to SatMAE-compatible EuroSAT classification, and Linhe uses Esri-derived supervisory labels rather than independent manual ground truth. The completed EuroSAT capacity sweep supports the capacity-boundary interpretation within the tested Prithvi setting, and the SatMAE-compatible second-backbone result supports bounded two-backbone consistency on EuroSAT. The same boundary discipline applies to deployment: Paper12 is not a validated ArcGIS replacement until same-area manual labels, ArcGIS reference outputs, Paper12 checkpoint-backed outputs, and paired metrics are available. It still does not establish a universal GeoFM ranking, because only one additional backbone and one second-backbone dataset have been tested. Despite these limits, the benchmark provides a reproducible stress-test protocol for GeoFM adaptation:

inspect the architecture before injecting PEFT modules, control the channel bridge, compare capacity as well as method family, and report per-class diagnostics when domain shift is severe.

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## Appendix A. Full Per-Seed Results

This appendix reports the per-seed values that underlie the aggregated tables in the main paper. All EuroSAT runs use 50 epochs; BigEarthNet-S2 runs use 15 epochs. Seeds are $\{42, 123, 456\}$ throughout.

### *Appendix A.1. EuroSAT Per-Seed Overall Accuracy*

Table A.8: EuroSAT per-seed overall accuracy. Each cell is a single training run.

| Method | Modality | seed=42 | seed=123 | seed=456 |
|---|---|---|---|---|
| Linear Probe | s2_full | 0.6589 | 0.6543 | 0.6572 |
| | rgb | 0.5570 | 0.5594 | 0.5519 |
| | rgb_sar | 0.5544 | 0.5531 | 0.5498 |
| | gf2 | 0.6452 | 0.6457 | 0.6559 |
| | sar_only | 0.3870 | 0.3833 | 0.3913 |
| BitFit | s2_full | 0.6996 | 0.7026 | 0.7035 |
| | rgb | 0.6076 | 0.6089 | 0.6067 |
| | rgb_sar | 0.6072 | 0.6039 | 0.6052 |
| | gf2 | 0.7004 | 0.7019 | 0.6998 |
| | sar_only | 0.4470 | 0.4480 | 0.4524 |
| LoRA (r=8) | s2_full | 0.6604 | 0.6548 | 0.6576 |
| | rgb | 0.5546 | 0.5596 | 0.5546 |
| | rgb_sar | 0.5557 | 0.5530 | 0.5504 |
| | gf2 | 0.6444 | 0.6480 | 0.6563 |
| | sar_only | 0.3913 | 0.3831 | 0.3900 |
| Houlsby | s2_full | 0.8189 | 0.8207 | 0.8241 |
| | rgb | 0.7294 | 0.7278 | 0.7244 |
| | rgb_sar | 0.7389 | 0.7361 | 0.7387 |
| | gf2 | 0.8226 | 0.8198 | 0.8178 |
| | sar_only | 0.6111 | 0.6161 | 0.6191 |
| LoRA Split-QKV | s2_full | 0.7093 | 0.7080 | 0.7037 |
| | rgb | 0.4969 | 0.4922 | 0.5035 |
| GeoAdapter v2 | s2_full | 0.5494 | 0.4607 | 0.4448 |
| | rgb | 0.3878 | 0.3956 | 0.3659 |
| Full Fine-Tuning | s2_full | 0.8685 | — | — |

*Appendix A.2. BigEarthNet-S2 Per-Seed mAP*

Table A.9: BigEarthNet-S2 per-seed validation mAP (19-class multi-label).

| Method | Modality | seed=42 | seed=123 | seed=456 |
|---|---|---|---|---|
| Linear Probe | s2_full | 0.3578 | 0.3606 | 0.3548 |
| | rgb | 0.3344 | 0.3343 | 0.3363 |
| LoRA (r=8) | s2_full | 0.3582 | 0.3606 | 0.3545 |
| | rgb | 0.3346 | 0.3350 | 0.3363 |
| Houlsby | s2_full | 0.4927 | 0.4815 | 0.4979 |
| | rgb | 0.4531 | 0.4583 | 0.4559 |

## Appendix B. Trainable Parameter Budgets

Table B.10 reports the exact trainable parameter count per method, isolated from the frozen Prithvi-100M backbone. The classification head accounts for 7,690 parameters on EuroSAT (10 classes) and 14,611 on BigEarthNet-S2 (19 classes).

Table B.10: Trainable parameter counts (head included).

| Method | EuroSAT | BigEarthNet |
|---|---:|---:|
| Linear Probe | 7,690 | 14,611 |
| GeoAdapter v2 | 7,873 | — |
| BitFit | 110,602 | — |
| LoRA (r=8) | 155,146 | 162,067 |
| LoRA Split-QKV | 597,514 | — |
| Houlsby (d=64) | 1,197,322 | 1,204,243 |
| Full Fine-Tuning | 86,244,874 | — |

## Appendix C. Training Curves

Figure C.7 compares the training loss trajectories of full fine-tuning and split-QKV LoRA on EuroSAT s2_full. Full fine-tuning converges to a substantially lower loss, consistent with its higher final accuracy. Figure C.8 shows the BigEarthNet-S2 loss curves for the three methods evaluated on that dataset; Houlsby separates from the LoRA/linear-probe cluster early in training.

## Appendix D. Hyperparameters and Compute

*Optimizer and schedule..* All runs use AdamW Loshchilov and Hutter (2019) with cosine annealing Loshchilov and Hutter (2017). Weight decay is $10^{-4}$. The PEFT-parameter learning rate is $10^{-3}$ on EuroSAT and $5 \times 10^{-4}$ on BigEarthNet-S2; the classification head receives $5 \times 10^{-3}$ throughout. Full fine-tuning uses a single learning rate of $5 \times 10^{-5}$.

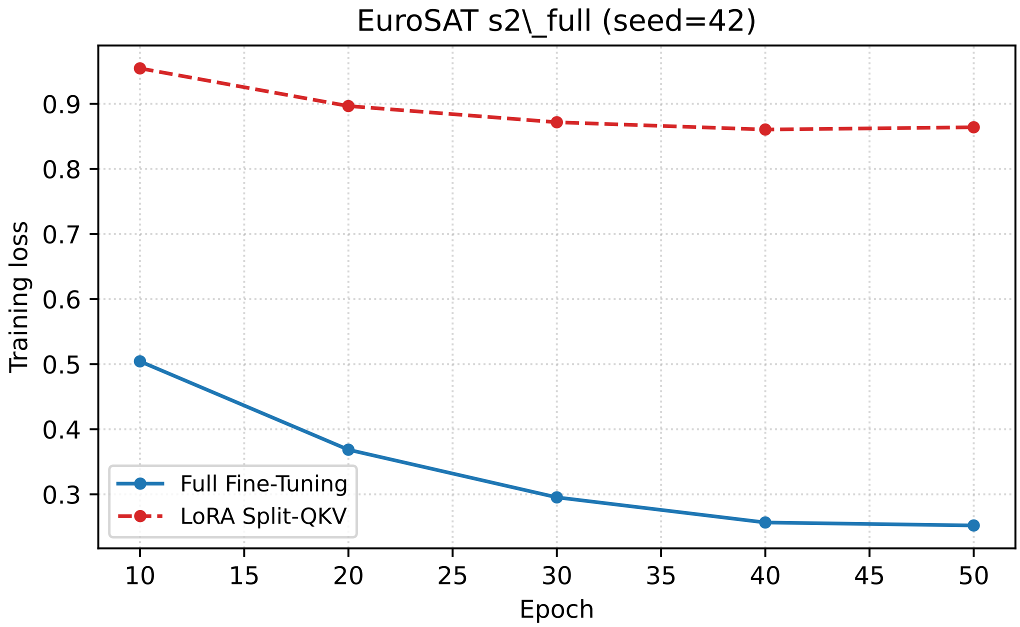


Figure C.7: Training loss on EuroSAT s2_full (seed=42). Full fine-tuning reaches a much lower loss floor than split-QKV LoRA.

*Batching..* Standard PEFT runs use batch size 64. Full fine-tuning uses batch size 32 to fit within 40 GB of A100 memory. BigEarthNet uses batch size 64 for all methods.

*Compute..* All experiments run on a single NVIDIA A100 40 GB on Google Colab Pro+. Wall-clock time per run averages 19–23 minutes on EuroSAT and 28–34 minutes on BigEarthNet (15 epochs over the 10K subsample). Total compute for the 100 reported runs is approximately 38 GPU-hours.

*Data preparation..* EuroSAT patches are resized from $64 \times 64$ to $224 \times 224$ and standardized per-band against statistics computed on the training split. BigEarthNet patches are decoded from the GeoTIFF format, resized to $120 \times 120$ and standardized identically. Channel zero-padding is applied at the dataloader level to bridge any modality with fewer than 6 channels to Prithvi's 6-channel input expectation.

## Appendix E. LoRA Rank Sensitivity

To test whether the split-QKV LoRA result is merely limited by an underpowered rank choice, we ran a completed 30-run capacity sweep on EuroSAT s2_full. Split-QKV LoRA used $r \in \{4, 8, 16, 32, 64\}$ with trainable parameter counts 302,602, 597,514, 1,187,338, 2,366,986, and 4,726,282. The mean overall accuracies (3 seeds each) are $0.7089 \pm 0.0018$ ($r = 4$), $0.7070 \pm 0.0029$ ($r = 8$), $0.7069 \pm 0.0013$ ($r = 16$), $0.7064 \pm 0.0017$ ($r = 32$), and $0.7078 \pm 0.0022$ ($r = 64$). Figure 1 in the main paper visualizes the original rank subset; the raw 30-row sweep is released as `paper12_results/peft_capacity_sweep.json`.

The result is qualitatively stable: increasing the LoRA rank does not produce monotonic gains. The matched Houlsby sweep shows the opposite pattern: $d = 8$, 16, 32, and 64 reach

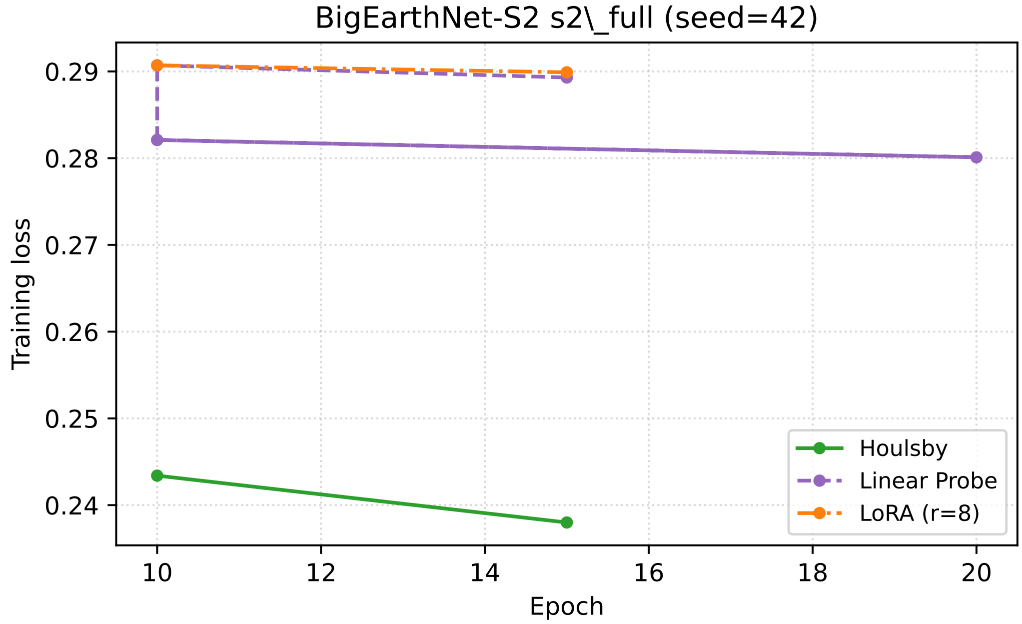


Figure C.8: Training loss on BigEarthNet-S2 s2_full (seed=42). Houlsby separates from LoRA and linear probing by epoch 10.

0.8644 ± 0.0032, 0.8815 ± 0.0033, 0.8932 ± 0.0010, and 0.9007 ± 0.0022 OA with 164,458, 312,010, 607,114, and 1,197,322 trainable parameters. The largest LoRA model therefore remains below the smallest Houlsby adapter despite using far more trainable parameters. This makes the post-fix LoRA ceiling much harder to dismiss as a hyperparameter or budget artifact. Within the explored range, the limitation appears architectural rather than rank-bound.

## Appendix F. LoRA Implementation Details and the Fused-QKV Trap

This section documents the exact code-level cause of the LoRA failure observed in the main paper, since reproducing or refuting our negative result requires reproducing the same insertion pattern.

Prithvi-100M's attention block is implemented as a `torch.nn.MultiheadAttention` module. In PyTorch's reference implementation, the query, key, and value projections are stored as a single tensor:

```
attn.in_proj_weight        # shape [3*D, D]
attn.in_proj_bias          # shape [3*D]
attn.out_proj.weight       # shape [D, D]
attn.out_proj.bias         # shape [D]
```

A naive LoRA insertion that iterates `module.named_modules()` and wraps every `nn.Linear` child therefore touches only `out_proj`: the actual Q, K, V projections are not children of `MultiheadAttention` and are silently skipped. Training proceeds, loss decreases, and accuracy settles a few hundredths above linear probing — which is exactly what we observe.

Our split-QKV variant resolves this by replacing each `MultiheadAttention` with three explicit `nn.Linear` projections initialized from the corresponding slices of `in_proj_weight`/`in_proj_bias`, then injecting LoRA into each of `q_proj`, `k_proj`, `v_proj`, and `out_proj`. The forward pass is rewritten to call each projection separately and re-combine the outputs through the standard scaled dot-product attention. We verified numerical equivalence on a small input batch before training begins.

We emphasize that this is not a bug in any specific PEFT library — it is a direct consequence of how PyTorch fuses attention projections by default. Any LoRA implementation that targets `nn.Linear` child modules will silently miss them on Prithvi-100M and other Vision Transformer codebases that build on `nn.MultiheadAttention`.

## Appendix G. LandCover.ai Per-Seed Segmentation Results

Table G.11: LandCover.ai per-seed mIoU. Each cell is a single training run (50 epochs, batch size 8, RGB modality).

| Method | Trainable Params | seed=42 | seed=123 | seed=456 |
|---|---|---|---|---|
| Linear Probe | 4,614 | 0.2508 | 0.2513 | 0.2507 |
| LoRA (r=8) | 152,070 | 0.2511 | 0.2511 | 0.2509 |
| Houlsby | 1,194,246 | 0.6392 | 0.6394 | 0.6444 |

The LoRA and linear-probe seed spreads agree to the fourth decimal place; the Houlsby spread spans 0.0052, dominated by the seed=456 run. No seed reverses the overall ordering.

### *Appendix G.1. Linhe LULC Per-Seed Results*

Table G.12 reports per-seed mIoU for the methods reported in Section 6.

Table G.12: Per-seed mIoU on Linhe LULC 6-class segmentation (geographic scene-level split, Esri 2022 labels).

| Method | Trainable Params | seed=42 | seed=123 | seed=456 |
|---|---|---|---|---|
| Linear Probe | 4,614 | 0.1778 | 0.1796 | 0.1752 |
| BitFit | 107,526 | 0.1977 | 0.2003 | 0.1946 |
| LoRA (r=8, split-QKV) | 152,070 | 0.1780 | 0.1788 | 0.1748 |
| Houlsby | 1,194,246 | 0.2966 | 0.2971 | 0.2860 |
| GeoAdapter | 4,756 | 0.2514 | 0.2751 | 0.2681 |

The Linhe seed spreads reproduce two qualitative patterns from the public benchmarks. First, LoRA's per-seed values $\{0.1780, 0.1788, 0.1748\}$ are statistically indistinguishable from linear probing's $\{0.1778, 0.1796, 0.1752\}$ to the fourth decimal place — the same diagnostic signature of LoRA collapse we observed on EuroSAT (Section 4) and LandCover.ai (Appendix Appendix A). Second, seed variance is small relative to the Houlsby–linear-probe gap: the largest per-seed Houlsby reading (0.297) exceeds the largest linear-probe reading (0.180) by more than 17 mIoU points, so no choice of seed reverses the ordering. GeoAdapter shows the largest seed spread (0.024), consistent with its higher variance reported on EuroSAT.

*Appendix G.2. LoveDA Cross-Domain Per-Seed Results and Collapse Diagnostics*

Table G.13 reports the 30 per-seed mIoU values that underlie Section 6.4.

Table G.13: Per-seed mIoU on LoveDA cross-domain 7-class segmentation. Each cell is one independent training run.

| | | Urban $\rightarrow$ Rural | | | Rural $\rightarrow$ Urban | | |
|---|---|---|---|---|---|---|---|
| Method | Params | seed=42 | seed=123 | seed=456 | seed=42 | seed=123 | seed=456 |
| Linear Probe | 5,383 | 0.0857 | 0.0859 | 0.0857 | 0.0949 | 0.0955 | 0.0955 |
| BitFit | 127,495 | 0.0850 | 0.0852 | 0.0849 | 0.0984 | 0.0986 | 0.0982 |
| LoRA (r=8) | 152,839 | 0.0856 | 0.0854 | 0.0855 | 0.0954 | 0.0951 | 0.0953 |
| Houlsby | 1,195,015 | 0.1370 | 0.1397 | 0.1387 | 0.2097 | 0.2130 | 0.2132 |
| GeoAdapter | 5,525 | 0.0854 | 0.0858 | 0.0878 | 0.0990 | 0.1013 | 0.1066 |

The seed dispersion for the four small-PEFT methods is at the fourth decimal place ($\sigma \leq 0.0014$), confirming that the cluster near 0.085 / 0.095 is not a seed artifact. Houlsby's seed dispersion is similarly tight ($\sigma = 0.0014$ U$\rightarrow$R, $\sigma = 0.0019$ R$\rightarrow$U), so the per-method ordering reproduces under all $3 \times 3 = 9$ seed pairings.

*Collapse-vs-undertraining diagnostic..* To distinguish "small PEFT collapses to majority-class prediction" from "small PEFT learns weak signal that mIoU under-reports," we extended the streaming evaluator to record per-class IoU and the predicted-pixel histogram alongside mIoU, and we ran a single-seed sweep on the U$\rightarrow$R direction with the LoRA budget enlarged 5$\times$ in learning rate ($\mathrm{lr}_{\mathrm{peft}}$=$5\times10^{-4}$) and 2$\times$ in epochs (60). The diagnostic config and the extended evaluator are released as `geoadapter/bench/configs/loveda_lulc_u2r_diag.yaml` and `geoadapter/engine/evaluator.py` respectively. Table G.14 reports the per-class IoU vector and the dominant-prediction share recovered by the diagnostic sweep.

Table G.14: LoveDA U$\rightarrow$R diagnostic sweep (single seed, 60 epochs, $\mathrm{lr}_{\mathrm{peft}}$=$5\times10^{-4}$). Per-class IoU columns 0–6 follow the LoveDA taxonomy: 0 background, 1 building, 2 road, 3 water, 4 barren, 5 forest, 6 agriculture. "dom%" is the share of validation pixels assigned to the dominant predicted class.

| Method | Params | 0 | 1 | 2 | 3 | 4 | 5 | 6 | dom% |
|---|---|---|---|---|---|---|---|---|---|
| LoRA r=8 diag | 152,839 | 0.397 | 0.201 | 0.000 | 0.001 | 0.010 | 0.021 | 0.000 | 87.8 |
| LoRA r=16 diag | 300,295 | 0.396 | 0.201 | 0.000 | 0.001 | 0.010 | 0.022 | 0.000 | 87.7 |

The signature is consistent across LoRA ranks: both diagnostic runs recover the building class (IoU $\approx$ 0.20) at the cost of background IoU dropping from the all-bg ceiling of 0.60 to 0.40, while only background and building exceed 0.05 IoU and the other five classes stay near zero. The dominant-prediction share falls from the all-bg ceiling of $\approx$100% to $\approx$88%, confirming that the model is no longer in pure majority-class collapse, but the non-degenerate class diversity it has learned is still limited to two classes. Doubling LoRA's rank and quadrupling its trainable-parameter budget over the protocol-default schedule yields a +0.0046 mIoU gain (best diagnostic mIoU 0.0901 vs. Phase 2 mean 0.0855), which we read as ruling out undertraining as the explanation for the small-PEFT plateau: the optimization budget is not the bottleneck, the per-method parameter budget is. We retain the term "capacity threshold" for the gap between sub-$10^6$ adapters (two non-degenerate classes only) and Houlsby at $\approx 1.2 \times 10^6$ parameters (multi-class transfer beyond the small-PEFT plateau).

## Appendix H. Reproducibility

Source code, training logs, raw experiment JSON files, and the figure-generation script are released under the project repository. Each experiment record contains the method name, modality, random seed, trainable parameter count, and metric values. Aggregated tables in the main paper can be reconstructed from these records using `paper12/scripts/make_figures.py` and `summary.csv`. We additionally provide `paper12_results/review_audit_summary.json`, a deterministic derived file that records the exact capacity-sweep, channel-bridge, LoveDA full-finetuning, LoveDA diagnostic, model-scope, label-source, and decoder-capacity checks used to support the bounded claims in the Discussion.